Enrollment No._____

MID-TERM EXAMINATION
B.Tech (ECE-AI) 1st semester
(January, 2023)
OFF LINE mode

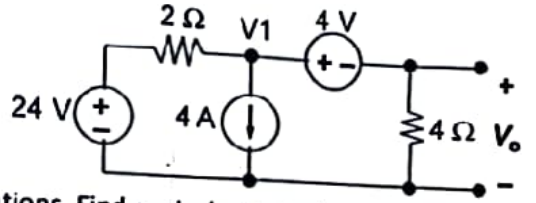
Subject Code: BEC-110
Time : 1 ½Hours
Subject: Basic Electrical Engineering
Maximum Marks : 30

Q1 Attempt all Parts **(2.5*4)**

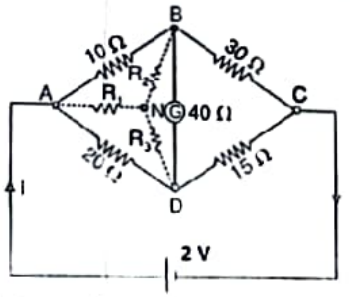
(a) StateSuperpositiontheorem with example.
(b) A 50Ωresistor is in parallel with 100Ωresistor. Current in 50Ωresistor is 7.2 A. How will you add a third resistor and what will be its value of the line-current is tobe its value if the line-current is to he 12.1 A?
(c) Differentiate between Q-factor, form factor and power factor of a series resonant circuit.
(d) A non-inductive resistance of 10 Ω is connected in series with an inductive coil across 200V, 50Hz AC supply. The current drawn by the series combination is 10A. The resistance of the coil is 2Ω. Determine inductance and power factor of the coil.

Q2 (Attempt any Two Parts) **(5,5)**

(a) Prove maximum power transfer theorem. Determine V_o and P_{MAX} of the circuit shown below:



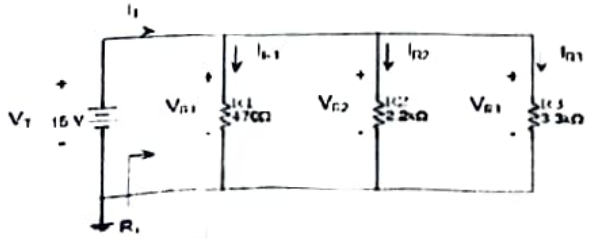
(b) Explain Y/Δ transformations. Find equivalent resistance across AC in figure below:



(c) ExplainThévenin's theorem utilizing nulling effect with example.

Q3 (Attempt any Two Parts) **(5,5)**

(a) A coil of resistance 15Ω & Inductance 0.05H is connected in parallel with a non-Inductive resistance of 20Ω. Design the respective circuit and find (I)current in each branch (ii)Total current (iii) Phase angle when a combination has applied a voltage of 200V at 50Hz (iv)Total power consumed.
(b) Consider the circuit shown below. Find(i) total resistance (R_T)(ii) Voltage across each component (V_T,V_{R1}, V_{R2}&V_{R3}) (iii) Current across each component (I_T,I_{R1}, I_{R2}&I_{R3}). Verify Kirchhoff's Current law using the results.



(c) The equation of an alternating current *i* is *i*=42.42 sin628t. find (i)max value (ii)frequency (iii) effective value (iv) average value (v)form factor